UK Maths Trust

Primary Kangaroo

Thursday 19 March – Thursday 26 March 2026

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England & Wales: Year 6 or below | *Scotland: P7 or below* | *Northern Ireland: Year 7 or below*

Instructions

1. Do not open the paper until the invigilator tells you to do so.
2. Time allowed: **45 minutes**.
   No answers, or personal details, may be entered after the allowed time is over.
3. The use of blank or lined paper for rough working is allowed; **squared paper**, **calculators and measuring instruments are forbidden**.
4. **Use a B or an HB non-propelling pencil.** Mark at most one of the options A, B, C, D, E on the Answer Sheet for each question. Do not mark more than one option.
5. **Do not expect to finish the whole paper in the time allowed.** The questions in this paper have been arranged in approximate order of difficulty with the harder questions towards the end. You are not expected to complete all the questions during the time. You should bear this in mind when deciding which questions to tackle.
6. **Scoring rules:**
   1 mark is awarded for each correct answer.
   You will not lose marks for getting answers wrong.
7. **Your Answer Sheet will be read by a machine.** Do not write or doodle on the sheet except to mark your chosen options. The machine will read all black pencil markings even if they are in the wrong places. If you mark the sheet in the wrong place, or leave bits of eraser stuck to the page, the machine will interpret the mark in its own way, or reject the answer sheet.
8. **The questions on this paper are designed to challenge you to think, not to guess.** You will gain more marks, and more satisfaction, by doing one question carefully than by guessing lots of answers. This paper is about solving interesting problems, not about lucky guessing.

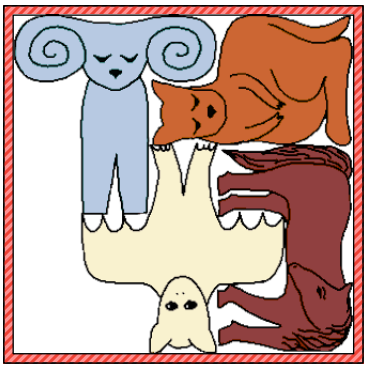
Enquiries about the Primary Kangaroo should be sent to:

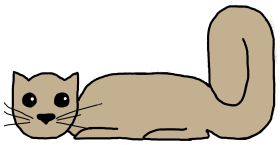
challenges@ukmt.org.uk

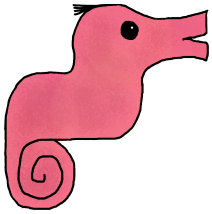
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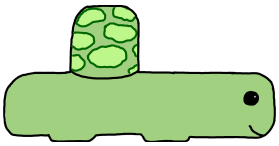
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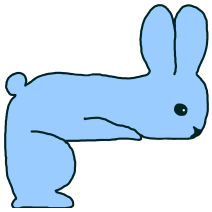
1. Which animal should Lisa use to fill the square on the right?

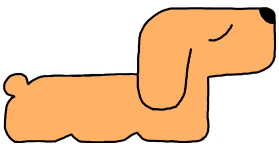


A 

B 

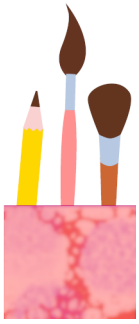
C 

D 

E 

2. Anita wants to make up some small boxes containing a pencil and one of each type of paint brush.

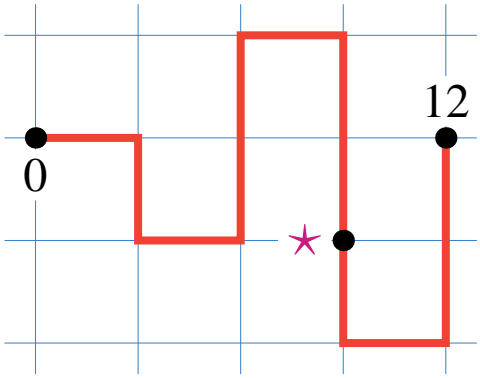




How many small boxes can Anita make from the larger box shown above?

A 2 B 3 C 4 D 5 E 6

3. Alina has a red number line that runs from 0 to 12. She bends it out of shape and places it on a grid, as shown below.



What number is at the ★ ?

A 5 B 6 C 7 D 8 E 10

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4. Drew has 5 pencils.

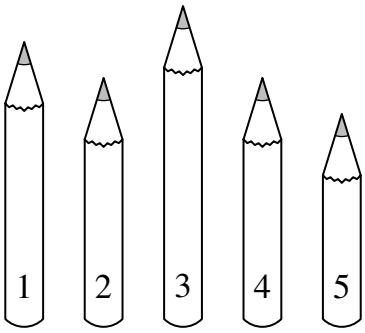
The shortest one is blue.

The green and red pencils are the same length.

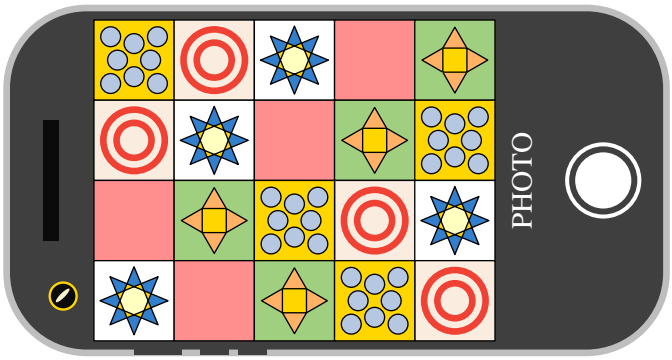
The purple pencil is shorter than the yellow one.

Which pencil is the purple one?

A 1 B 2 C 3 D 4 E 5

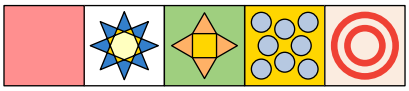


5. A floor is made of five different tiles. The tiles are laid in a repeating pattern. Evan takes a picture of part of the floor on his phone, as shown below.

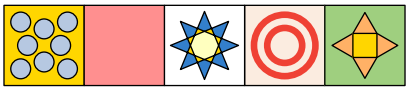


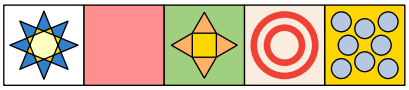
Which of the following is the repeating pattern of the five tiles?

A 

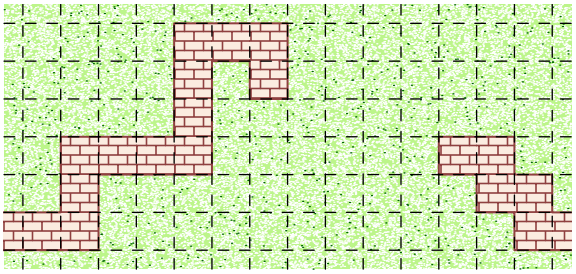
B 

C 

D 

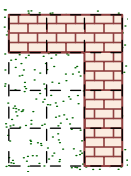
E 

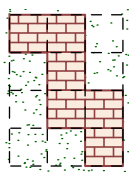
6. The path shown below needs an extra section to join its two parts together.

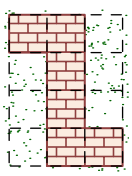


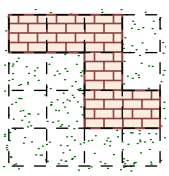
A section may **not** be turned and should share an edge at either end with the existing parts of the path.

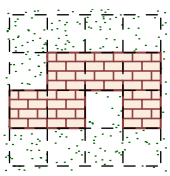
Which section will complete the path?

A 

B 

C 

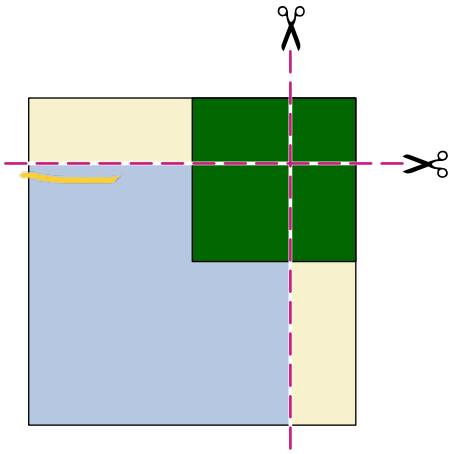
D 

E 

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7. Dan takes a cream-coloured square of paper.

He places a blue square on top of it, and then places a green square on top of that, covering a corner of the blue square and part of the cream square, as shown on the right.



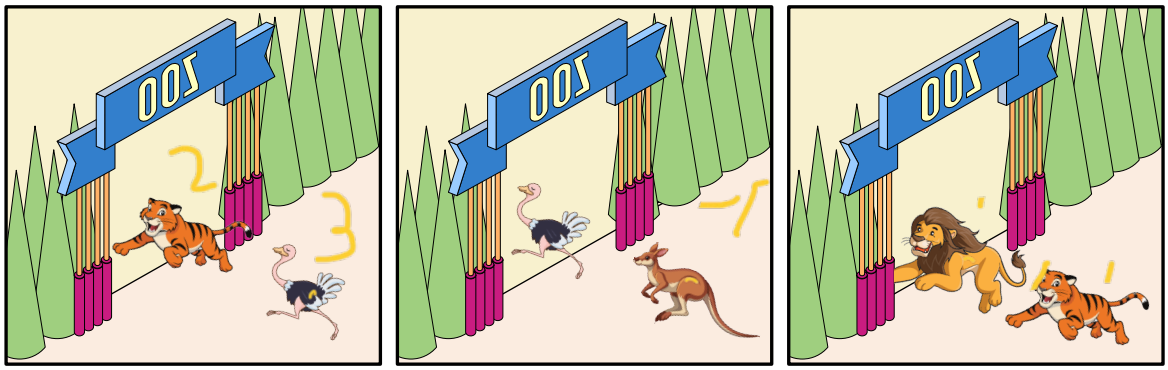
Then he cuts the pile of paper along the two lines shown.

How many of the pieces that he ends up with are squares?

A 3 B 4 C 5 D 7 E 9

8. The three photographs below show four animals escaping from the zoo:

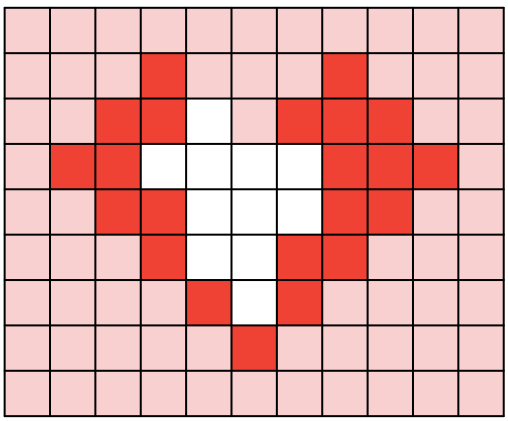
a tiger  , a kangaroo  , a lion  and an ostrich  .



In which order did they escape through the gate?

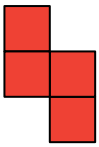
A lion, ostrich, tiger, kangaroo
B tiger, ostrich, kangaroo, lion
C ostrich, kangaroo, lion, tiger
D tiger, kangaroo, lion, ostrich
E lion, tiger, ostrich, kangaroo

9. Beth is making a heart-shaped puzzle for her mother, as shown on the right.



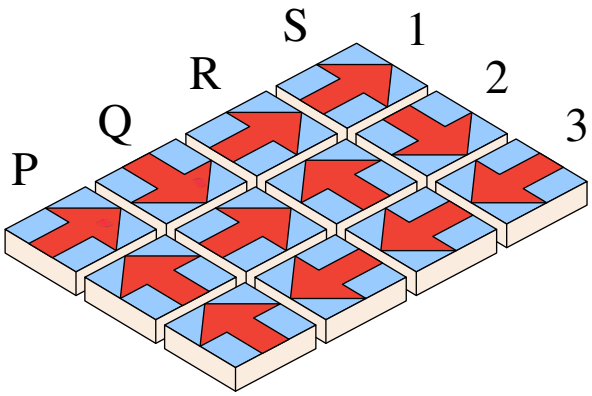
She intends to use four of the following pieces to finish it.

Which of these pieces does she have to leave out?

A 
B 
C 
D 
E 

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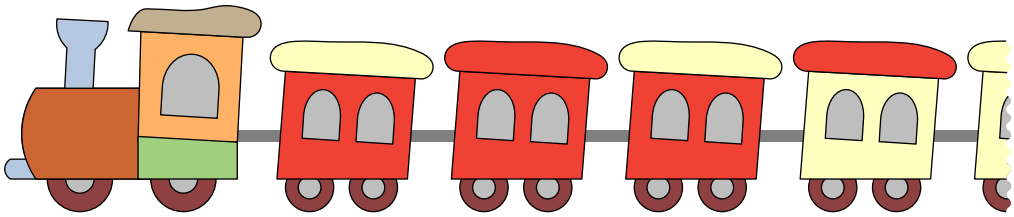
10. Each tile in the grid below has an arrow. The arrow shows the direction in which Indira must walk from one tile to the next.



From which tile should Indira start in order to walk over **all** of the tiles exactly once?

A P1 B Q1 C R2 D S2 E S3

11. A train has an engine that pulls 18 coaches.
The picture below shows the engine and the first four coaches of the train.



The colour of each coach goes in this repeating sequence:

red, red, red, yellow, yellow, blue.

Each coach has a roof and the roofs are coloured according to the repeating sequence:

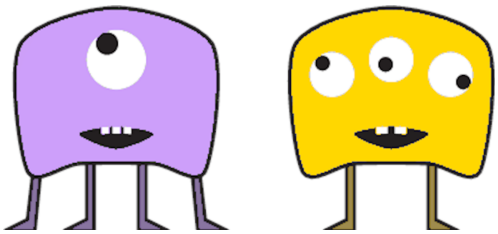
yellow, red, yellow, red.

How many of the 18 coaches are red with a red roof?

A 1 B 3 C 4 D 6 E 8

12. In a village, there are two types of monsters:

- monsters with 1 eye and 4 legs,
- monsters with 3 eyes and 2 legs.



Altogether the monsters have 9 eyes and 16 legs.

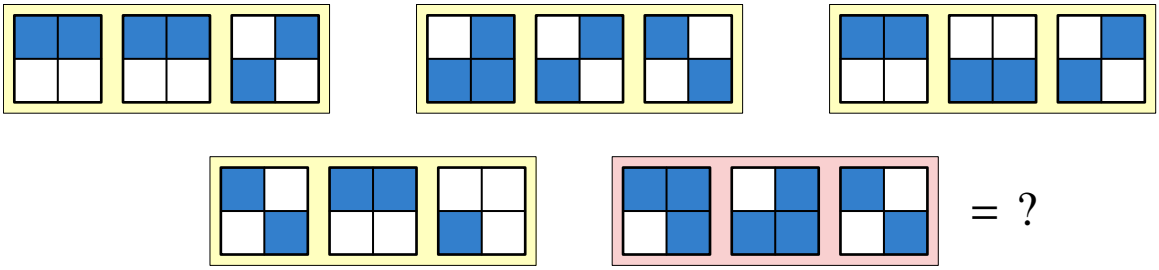
How many monsters with 1 eye are there?

A 1 B 2 C 3 D 4 E 5

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13. The five rows of patterns shown below each represent one of the three-digit numbers

183, 451, 521, 872, and 882,



though the five numbers are not necessarily in that order.

Each digit from 0 to 9 has its own unique pattern.

Which of the numbers is represented by the pattern with the question mark?

A 183 B 451 C 521 D 872 E 882

14. Dela, Ela, and Fela go to an ice cream shop.

One of the girls buys strawberry ice cream  , another buys vanilla ice cream  ,

and another buys chocolate ice cream  .

Each girl buys a different number of scoops: 1, 2, or 3.

We know that:

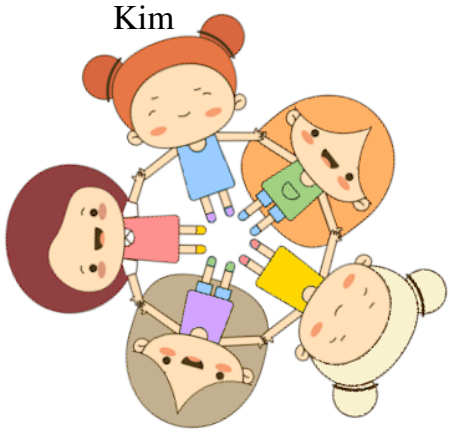
- Dela does not buy strawberry ice cream and chooses the smallest amount,
- Ela does not buy chocolate ice cream,
- The chocolate ice cream is the largest.

Which ice cream does Fela buy?

A  B  C  D  E 

15. Five children, Jun, Kim, Lea, Mai and Nao, lie down and hold hands to make a circle.

- Jun holds Kim's left hand,
- Mai does not hold Lea's hand,
- Lea holds Nao's right hand.



Who is Mai holding hands with?

A Nao and Kim B Jun and Kim C Lea and Kim
D Jun and Nao E Lea and Nao

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16. Rosie wishes to write the numbers 2, 0, 2, 6 in the square grid shown on the right, so that in each row and in each column there are exactly two 2s, one 0 and one 6.

2			
	0		?
		2	
	?		6

She has already written some of the numbers.

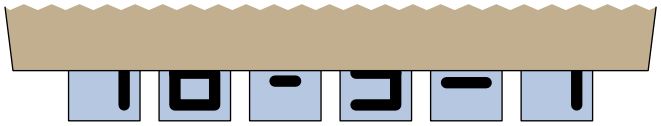
When Rosie has finished, what will be the sum of the numbers in the squares with the question marks?

A 0 B 2 C 4 D 6 E 8

17. Jack has 13 cards, each of which has a digit or an arithmetic symbol (+, − or =):



He chose six of these cards and, without rotating any of them, he placed them together to form a valid calculation. Then he placed a large card over the top half of them:



What is the sum of the four digits Jack used?

A 19 B 21 C 23 D 25 E 33

18. Three girls Non, Ida and Zoe, each have a box of sticks.

One of the girls has a box of 1 cm sticks, another of the girls a box of 2 cm sticks, and another a box of 3 cm sticks; however, we do not know which girl has which box.

Non places one of her sticks flat on the ground.

Then each girl places one of her sticks to the right of the previous stick in the order Ida, Zoe, Non, Ida, Zoe, Non and so on, making a long straight line.

They stop when the line is exactly 50 cm long.

Which two sticks are at the beginning and at the end of the line?

A 1 cm … 2 cm

B 2 cm … 3 cm

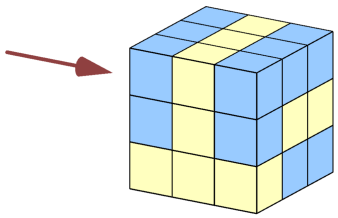
C 3 cm … 1 cm

D 3 cm … 3 cm

E 2 cm … 2 cm

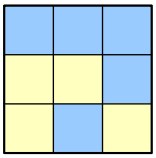
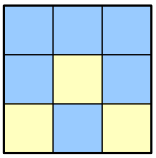
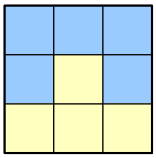
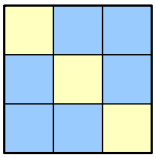
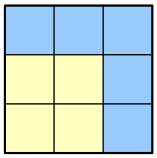
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19. A big cube, made out of 16 blue cubes and 11 yellow cubes, all of the same size, is shown below.

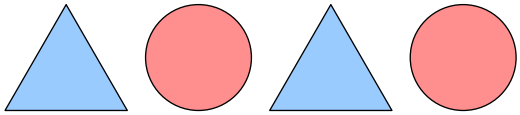


Markiyan is looking towards the cube from its left, in the direction shown by the arrow.

Which of the following could be the face that he sees?

A 
B 
C 
D 
E 

20. There are four buttons in a row on a touch screen. Two of them show circles and two show triangles.



When a button is pressed, its shape and the shapes of any buttons directly next to it change.
A circle becomes a triangle and a triangle becomes a circle.

What is the smallest number of button presses needed for the screen to show four circles?

A 2 B 3 C 4 D 5 E 6

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